

**BEng (Hons) Software Engineering**

**SWE6013 Enterprise System  
Development**

**MODULE GUIDE**

**2025/2026**

**Semester 3 Cohort 13,14**

**Level HE6**

## Contents

|                                                                               |   |
|-------------------------------------------------------------------------------|---|
| 1. Key Information .....                                                      | 2 |
| 2. Module Communications.....                                                 | 2 |
| 3. Module Description .....                                                   | 2 |
| 4. Learning and Teaching .....                                                | 2 |
| 5. Learning Outcomes and Assessments .....                                    | 3 |
| 6. Assessment Deadlines.....                                                  | 4 |
| 7. Formative Assessment .....                                                 | 4 |
| 8. Summative Assessment Feedback .....                                        | 4 |
| 9. Module Calendar.....                                                       | 5 |
| 10. Indicative Reading – Essential and Recommended .....                      | 5 |
| 11. Guidelines for the Preparation and Submission of Written Assessments..... | 6 |
| 12. Late work.....                                                            | 6 |
| 13. Extensions .....                                                          | 7 |
| 14. Procedures for Other Assessments .....                                    | 7 |
| 15. Academic Misconduct.....                                                  | 7 |
| 16. Assessments.....                                                          | 7 |

## 1. Key Information

|                                 |                                                                                                                                                                                                                        |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module Tutor                    | Ms Renuka Nyayadhish                                                                                                                                                                                                   |
| Module Tutor Email              | r.nyayadhish@bolton.ac.uk                                                                                                                                                                                              |
| Other Contact Methods           | Via Moodle                                                                                                                                                                                                             |
| Availability                    | Please allow 48 hours for any response                                                                                                                                                                                 |
| Weblink to Moodle Class         | <a href="https://moodle.greatermanchester.ac.uk/course/view.php?id=38372">https://moodle.greatermanchester.ac.uk/course/view.php?id=38372</a>                                                                          |
| Class Zoom Link                 | <a href="https://us06web.zoom.us/j/84136883689?pwd=s4cfSKmjLjSX2ZJ1wzaX2kLgFqLsN.1">https://us06web.zoom.us/j/84136883689?pwd=s4cfSKmjLjSX2ZJ1wzaX2kLgFqLsN.1</a><br><br>Meeting ID: 841 3688 3689<br>Passcode: 634373 |
| Weblink to Module Specification | <a href="https://modules.bolton.ac.uk/SWE6013">https://modules.bolton.ac.uk/SWE6013</a>                                                                                                                                |

## 2. Module Communications

The Module Tutor's contact details are provided above. In addition to your personal email, you should check your **University of Greater Manchester email address** and the **Moodle** area dedicated to this module regularly as many module communications are channelled through these routes.

Your Module Tutor will normally aim to respond to your email messages within **2 full working days** of receipt.

## 3. Module Description

This module provides an opportunity to consolidate and extend your knowledge and understanding of enterprise-level software systems, recognising the complexity of designing, integrating and maintaining applications.

Drawing on both theoretical foundations and practical application, you will critically analyse enterprise development frameworks, architectural patterns and technologies to design and implement secure, scalable and maintainable solutions that address real-world business needs.

This module complements your other software engineering studies by strengthening your ability to align technical development with organisational goals, while developing your research, problem-solving and critical evaluation skills through the delivery of a substantial enterprise systems project.

## 4. Learning and Teaching

This module is delivered via online live sessions and asynchronous practical activities. Live sessions will include lectures, break-out room discussions, workshops and tutorials. All lectures

will be recorded to allow students to check/refresh learning. Any asynchronous practical activities should be completed by time specified.

As well as completing the asynchronous practical activities in your own time, you are also expected to carry out independent study including research and assessment activities.

In weeks 6,9 and 10 tutorials will take place which will provide you with the opportunity to discuss your progress on the module and receive feedback on formative assessments.

**When attending live sessions online, you are expected to:**

- keep your camera on to show engagement with the session (and agree with the tutor beforehand if this is not possible)
- write your full name in your Zoom profile in advance of the session
- find a quiet workspace where you are less likely be interrupted for the duration of session – and/or wear a headset to reduce background noise disruption
- ensure your appearance and surroundings are appropriate for the online classroom (or use an appropriate background picture)
- be prepared for the session and join punctually
- keep your microphone on mute when you are not speaking in the session to ensure background noise does not disrupt the session
- stay seated, be present and participate during the live session, as well as in breakout room discussions
- use the chat function to ask questions and raise points
- ensure what you write in the chat is relevant to the session and courteous
- use the hand up/wave icon if you wish to raise a point verbally.

You are expected to attend at least 80% of live sessions and engage with the module fully. If you are unable to attend a scheduled session or engage with a set activity due to factors outside your control, you should inform your tutor and email [info@wincedu.uk](mailto:info@wincedu.uk)

**Please note that attendance and engagement is taken into account when making Assessment Board decisions.**

## 5. Learning Outcomes and Assessments

| Module Learning Outcomes and Type                                                                                    | Assessment No. and Type                |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| LO1: Identify and appraise the role of patterns and frameworks in the design of enterprise application architecture. | Assessment No 1-Written Project Output |
| LO2: Critically analyse, select and synthesize existing technologies to develop an enterprise application.           | Assessment No 1-Written Project Output |
| LO3: Autonomously construct and plan an enterprise application.                                                      | Assessment No 1-Written Project Output |
| LO4: Examine the relationship between business goals and software development.                                       | Assessment No 1-Written Project Output |

|                                                                                                                                                |                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| LO5: Research and implement a number of software elements in order to meet the business/infrastructure needs of enterprise in a secure manner. | Assessment No 1-Written Project Output |
|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|

## 6. Assessment Deadlines

| Assessment item |                                                                                                            | Due Date   | Weight |
|-----------------|------------------------------------------------------------------------------------------------------------|------------|--------|
| 1               | <b>Formative: Written Project Output-Report</b>                                                            | 12/08/2026 | 100%   |
|                 | <b>Summative: Written Project Output - Enterprise system study (Report , Application and Presentation)</b> | 25/08/2026 |        |

## 7. Formative Assessment

Formative assessment is employed to support your learning on the module, allowing you to reflect on feedback on your progress from your tutors and peers. As well as the formal formative assessment outputs set out in Section 7, formative assessment includes asynchronous practical activities and workshop activities in the live sessions. Formative assessment does not directly contribute to the final module mark

## 8. Summative Assessment Feedback

Feedback on summative items of assessment can be formal (such as on a signed feedback form) or informal (such as advice from a tutor in a tutorial). Feedback is therefore not just your grade or the comments written on your feedback form, it is advice you get from your tutor and sometimes your peers about how your work is progressing, how well you have done, what further actions you might take.

We recognise the value of prompt feedback on work submitted. Other than in exceptional circumstances (such as might be caused by staff illness), you can expect your work to received provisional marks and feedback no more than 15 working days from the deadline date. However, please note that marks are not agreed until the Assessment Board has met and may therefore be subject to change.

Please take time to read or listen to your assessment feedback. This can be very useful in determining your strengths and key areas for development and can therefore help you improve on future grades.

## 9. Module Calendar

| Session No. | Date or Week Commencing | Topics Covered                                                                                                                                                                                       |
|-------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1           | 25-May-26               | Introduction to Enterprise Architecture                                                                                                                                                              |
| 2           | 1-Jun-26                | Enterprise architecture Details                                                                                                                                                                      |
| 3           | 8-Jun-26                | Introduction to Enterprise architecture frameworks<br><b>Assessment briefing</b><br>TOGAF framework<br>Zachman Framework<br>ITIL Framework<br>Identifying the business goals and required components |
| 4           | 15-Jun-26               | Zachman Framework<br>ITIL Framework<br>Selection of Framework                                                                                                                                        |
| 5           | 22-Jun-26               | Types of Software Architecture<br>Review on the Case study Business context architecture and required components analysis<br>System design – ER diagrams, Data Flow diagrams                         |
| 6           | 29-Jun-26               | REQUIREMENTS CATALOGUE -System design diagrams –<br>Use case Diagram                                                                                                                                 |
| 7           | 6-Jul-26                | Mapping Business goals and requirements                                                                                                                                                              |
| 8           | 13-Jul-26               | ER Diagram                                                                                                                                                                                           |
| 9           | 20-Jul-26               | System architecture and Design - DFD<br><b>Tutorial for system design</b>                                                                                                                            |
| 10          | 27-Jul-26               | <b>Project Planning</b>                                                                                                                                                                              |
| 11          | 3-Aug-26                | <b>Implementation Details</b>                                                                                                                                                                        |
| 12          | 10-Aug-26               | <b>Software testing</b>                                                                                                                                                                              |
| 13          | 17-Aug-26               | Assignment Formative Feedback                                                                                                                                                                        |
| 14          | 24-Aug-26               | Assignment Formative Feedback                                                                                                                                                                        |

## 10. Indicative Reading – Essential and Recommended

### Essential Reading

- Freeman, A. (2017) Pro ASP.NET Core MVC 2. New York: Apress
- Lankhorst, M. (2013) Enterprise architecture at work modelling, communication and analysis (3rd ed.). London: Springer
- Northwood, C. (2018) The Full Stack Developer. New York: Apress

### Other Recommended Reading

- Tatroe, K., MacIntyre, P(2020) Programming PHP: creating dynamic web pages.
- Fourth edition
- Nick, G., Nanik, T., Tolaram (2023) Becoming a Full Stack Go Developer: Packt Publishing
- Medium. (2021) 10 Software Architecture Patterns in Enterprise Software Development. (Online) Available from: <https://blog.devgenius.io/10-software-architecture-patterns-in-enterprise-software-development-fabacb5ed0c8>.
- Kravchenko, I. (2022) Enterprise Software Development. (Online) Available from: <https://diceus.com/enterprise-software-development-guide/>.

## 11. Guidelines for the Preparation and Submission of Written Assessments

1. Written assessments should be word-processed in Arial or Calibri Light font size 12. Each page should be numbered.
2. There should be a title page identifying the programme name, module title, assessment title, your student number, your marking tutor and the date of submission.
3. You should include a word-count on the assignment cover page (excluding reference list/bibliography, figures, tables and appendices).  
Where a word limit is specified, the following penalty systems applies:
  - Up to 10% over the specified word length = no penalty
  - 10 – 20% over the specified indicative word length = 5 marks subtracted (but if the assessment would normally gain a pass mark, then the final mark to be no lower than the pass mark for the assessment).
  - More than 20% over the indicative word length = if the assessment would normally gain a pass mark or more, then the final mark will be capped at the pass mark for the assessment.
4. **At the end of the assessment you should include a declaration of any software tools including Generative AI (GAI) applications that you used in developing and completing the assessment. The assessment brief will specify if and how you can use GAI applications in the assessment.**
5. All written work should be referenced using the standard University of Bolton referencing style– see: <https://libguides.bolton.ac.uk/resources/referencing/>
6. Unless otherwise notified by your Module Tutor, electronic copies of assignments should be saved as word documents and uploaded into Turnitin via the Moodle class area. If you experience problems in uploading your work, then you must send an electronic copy of your assessment to your Module Tutor via email BEFORE the due date/time.
7. Please note that when you submit your work to Moodle, it will automatically be checked for matches against other electronic information, as well as for hidden text characters and GAI generated text. You will be able to see similarity matches but not currently flags for hidden characters and AI-generated text. The outcomes of Turnitin reports may be used as evidence in an academic misconduct investigation (see Section 15).

## 12. Late work

Late work will be subject to the following penalties:

- Up to 7 calendar days late = 10 marks subtracted, but if the assignment would normally gain a pass mark, then the final mark to be no lower than the pass mark for the assignment.
- More than 7 calendar days late = This will be counted as non-submission and no marks will be recorded.

Late submission of assessments on refers and those which are graded Pass/Fail only, is not permitted unless an extension is approved.

### 13. Extensions

In the case of exceptional and unforeseen circumstances, an extension of up to 14 days after the assessment deadline may be requested using the standard University Extension Request Form. For approval there would need to be an explanation and evidence of relevant circumstances. Longer extensions for individual assessments and projects at the discretion of the Programme Leader.

Requests for extensions which take a submission date past the end of the module must be made using the *Special Circumstances* procedure

Some students with registered disabilities will be eligible for revised submission deadlines. Revised submission deadlines for disability adjustments do not require the completion extension request paperwork. However, students should request these in writing in advance.

Please note that the failure of data storage systems is not considered to be a valid reason for an extension. It is therefore important that you keep multiple copies of your work on different storage devices before submitting it.

### 14. Procedures for Other Assessments

Presentation and demo of the application will be conducted as per the schedule shared by the tutor.

### 15. Academic Misconduct

Academic misconduct may be defined as any attempt by a student to gain an unfair advantage in any assessment. This includes plagiarism, collusion, commissioning (contract cheating and unauthorised use of GAI) amongst other offences.

In order to avoid these types of academic misconduct, you should **ensure that all your work is your own and that sources and software applications are attributed.** You can also check originality through *Turnitin*.

Please note that penalties apply if academic misconduct is proven. See the following link for further details:

<https://greatermanchester.ac.uk/student-policy-zone/student-policies-2025-26>

### 16. Assessments



## Assessment Brief

|                                 |                         |
|---------------------------------|-------------------------|
| Assessment Number               | 1                       |
| Assessment Type (and weighting) | Portfolio (100%)        |
| Assessment Name                 | Enterprise system study |
| Assessment Submission Date      | 25/08/2026              |

### Learning Outcomes Assessed:

LO1: Identify and appraise the role of patterns and frameworks in the design of enterprise application architecture.

LO2: Critically analyse, select and synthesize existing technologies to develop an enterprise application.

LO3: Autonomously construct and plan an enterprise application.

LO4: Examine the relationship between business goals and software development.

LO5: Research and implement a number of software elements in order to meet the business/infrastructure needs of enterprise in a secure manner.

### Assessment Task:

#### Assessment Brief

**BookMyHotel** is a company which is based in Dubai and provide reservations for different hotels through diverse contracts. For hotel business sustainability, many hotels combine their services in a common platform. Data collected at booking systems sites could help to organize a competitive strategy for cost-effective hotel management system and hospitality industry development. Now the **BookMyHotel** Company needs a common platform that makes reservations for customers from different hotels in Asian and European countries. They found e-commerce web application will help their need and decided to have a website named **BookMyHotel .com**. Company's aim is to make a single accommodation booking website for different hotels. This company has made contracts with different hotels in Asian and European countries. Impressed by the technology offered, but in need of enterprise features at its scale—the site will handle approximately 300 room-night reservations a day on average. They are in need for the developers to design a well-functioning E-commerce web application. To find the requirements and expectations of the hotel customers **BookMyHotel** conducted an online survey in May 2025 and developed a case study. Based on the survey results **BookMyHotel** business requirements are discussed in the below case study. **BookMyHotel** has found the requirements based on the survey are satisfied by these 4 hotels and so considered for this common platform for bookings.

Consider the **BookMyHotel** Company has appointed you as the intern for their e-commerce web application development project. For the Software development project with documentation for the **BookMyHotel .com** the following tasks need to be accomplished. The intern needs to analyse the case study provided and find out the requirements. For an efficient and effective software project development, the intern needs to critically evaluate all the relevant enterprise system architectures, patterns, frameworks, and their roles in for an enterprise e-commerce web application. Also need to analyse all the software technologies and hardware for the successful implementation of the e-commerce web application. Based on the analysis identify the most suitable architecture, software and hardware infrastructure requirements for the development and deployment of the software. All the research and processes need to be documented and reported to the client.

### **Case study**

Tourism including hotel industry consists 10.6% of the world economy in a sense of GDP (Gross Domestic Product) as it is stated in United Nations World Tourism Organization (UNWTO) Report.

While customers are keen and want their future bookings to help rebuild communities around the world, they will expect a lot more value beyond the price tag want booking platforms to increase their transparency about cancellation policies, refund processes and insurance options. Doorstep delights offer both cost and time savings to which destinations and accommodations will respond by offering more historic and cultural heritage tours to entertain and attract more visitors, as well as competing for customers affections through original offerings such as guest chefs and bespoke cocktails.

Also, hotels need to be more cautious when it comes to health and safety. Hotels will have to work cohesively to set consistent standards to help keep customers safe, and with expectations heightened and businesses will need to work harder to gain customer's trust.

Hotels need to favour accommodations that have antibacterial and sanitizing products. Hotels need to be more sustainable, and customers will consequently need to adapt smart crowd management measures to appease customers visiting their hotels.

Customers' expectations for sustainable reservations in the future state that the customers like an economic incentive (e.g. tax breaks) for customers who opt for an eco-friendly booking in the future want online booking sites giving reward points for making sustainable choices to use for free extra perks and discounts. Expect booking sites to offer tips on how to be more sustainable. Customers are looking to online travel booking sites to use clearer signposting so they can easily identify when something is (not) sustainable. Customers expect that the industry must adapt to this sustainable mind-set for the long-term by offering more attractive off-season travel packages.

Travel platforms and places to stay will prioritise showcasing home office facilities and Wi-Fi speed to attract this new wave of digital nomads. Likewise, the world of corporate travel will see increasing demand for privacy, cleanliness and longer stays among those traveling for business, requiring alternative accommodations to seriously up their 'work-friendly' game.

While companies will undoubtedly reassess their approach to business travel in the future, workers will continue to maximise the trips they do take.

Considering the renewed emphasis on privacy, sufficient space and personal control over cleanliness and hygiene, it is not surprising that we will see customers look for accommodation 'closer to home'. Customers will opt to eat in more as opposed to eating out at restaurants, so a well-equipped kitchen will be essential for any vacation rental in a hotel.

Tech innovation will play a crucial role in rebuilding customers' confidence, and we will see the accelerated use of tech to adapt to the current pandemic issues. Tech will help us regain the spontaneity, confidence and ease of times past while at the same time help people travel safely and responsibly.

The software or platform need to provide unparalleled insights, to make it easier for everyone to experience the world; the software developed are there for the customers offering the widest choice, great value and the easiest experience from anywhere in the world. Reliance on technology will only continue to grow as technology proves its worth and becomes more ingrained into our travel experiences. The user interface needs to be better and should support various search options based on price, services and rooms. The payment methods should be different and need to be secure. The innovations will bring even more change, with enhanced online experiences influencing are excited about tech's potential to further personalize their reservations for accommodations uses the latest technologies to make customers feel safe. All security implications need to be considered for safe payment and refunds.

**BookMyHotel .com** guarantees the best prices for different hotels. Currently, the website will allow its customers to order from four famous hotels in Asia and Europe with various branches: The Marriott, the Hilton, the Hyatt and the Four Seasons. These hotels are five-star rated and have different types of rooms along with various facilities. **BookMyHotel** has found the requirements based on the survey are satisfied by these 4 hotels and so considered for this common platform for bookings.

Registrations need to be done by users for reservations. Users cannot login without registering. The customers should be able but not limited to view, search, reserve and cancel the available services from the hotels. Some of the common services of these hotels are bar and restaurant, rent of cars and motorbikes, sale of tickets for local trips with tourist guides, spa, etc. If they want to contact the company, they should be able to do that as well through contact form. Rates & Availability supports the calendar with room daily rates broken down into specific dates and types; rooms' occupancy and availability also are considered. Promotions allow to creating discount actions in addition to Rates & Availability. Reservations contain details of booking made by the past and future guests. Property supports description of hotel in respect with user interface reflected when guests come to your hotel page at the **BookMyHotel .com**. Opportunities should encourage hotels managers to provide some discounts to guests. Guest Reviews aggregates scores of written guests feedback and ratings.

The admin should be able to add, edit or remove hotels, their details along with reservations easily and be able to see the reservation details for a particular date. There is also a requirement of an Analytics Dashboard for Admin which aggregates the main performance overview including number of booked room nights, room revenue (the total amount paid by

guests), and average daily rate what is room revenue divided to room nights; also the Analytics Dashboard contains links to the main reports for each hotel. Since the **BookMyHotel** company is growing the number of hotels and services will increase in the future. Hence it should be scalable and efficient and make the company sustainable in the industry.

To get started on creating the web applications, it's highly important that the developer first focus on the requirements of **BookMyHotel** and then the system architecture and design of the website and how different functions connect with each other.

To make an effective enterprise system development, the analysis documentation, software development progress needs to be shown to the Client. Then the requirements of the **BookMyHotel .com** need to be properly analysed and mapped with the business goals. Then a project plan needs to be made for the development which includes all the resources, time and cost for the development.

Based on your evaluation of the technologies and requirements, make the proper design for the database and web services implementation. Then develop the database and web services as per the case study. Test the e-commerce web application satisfies all the business requirements. Since documentation is an integral part of the software development, the entire evaluation and development need to be documented as a report. A ReadMe file need to be developed which includes the steps to install and use the software. At the successful completion of the software project development, the software development process and the results need to be explained to the client as a 10-minute presentation along with a software demonstration.

### **Guidelines to complete the assignment**

1. Students have **CHOICE** to select enterprise framework and pattern. They can choose tools to create design diagrams and artefacts. They get choice to select tools/Language/Database for creating web application.
2. Analyse the case study in detail and progress every week for the development as per the requirements.
3. Critically evaluate the enterprise system architectures and frameworks and their role in enterprise system development.
4. Analyse and document the various requirements of the **BookMyHotel** Company for an e-commerce application for reservations of different facilities.
5. Make a project plan for the enterprise system development.
6. Make the proper mapping of the requirements based on business goals.
7. Evaluate the software and hardware infrastructures required for the **BookMyHotel** Company for the e-commerce web application.
8. Select the proper architecture for the successful implementation
9. Make the proper design for the successful enterprise infrastructure implementation

10. Implement the database for the hotels
11. Implement the services available for the hotels
12. Document the development and the results along with security concerns
13. Test the functionalities and document how it met the business goals
14. Present the findings and solutions to the audience with detailed explanation
- 15. The project report should be structured and referenced appropriately with critical reflection Enterprise system study and application**

**Declaration:** At the end of the assessment, you should also include a declaration of any software tools including Generative AI (GAI) applications that you used in developing and completing the assessment.

#### Use of Generative Artificial Intelligence (GAI) Applications in this Assessment

| AI Status         | Application                                                                                                                                                                                    | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Category B</b> | <p>GAI can be used to assist you with the assessment.</p> <ul style="list-style-type: none"> <li>- To explore the frameworks</li> <li>- To get information about the tools/app used</li> </ul> | <p>Grammar and/or spell checkers may be used to correct individual words and sentences.</p> <p>GAI can also be used for specific tasks, e.g.</p> <p>(1) to generate initial ideas which must then be subject to fact checking by authoritative sources, (2) to structure assessments, (3) to generate summary information which should then be put into your own words and referenced.</p> <p><u>Any GAI outputs which are presented as your own original work/skills and are not authorised and acknowledged will be assessed for academic misconduct.</u></p> |

#### Minimum Secondary Research Source Requirements:

**Level HE6** - It is expected that the Reference List will contain between **fifteen to twenty sources**. As a MINIMUM the Reference List should include **three refereed academic journals and five academic books**.

#### Specific Assessment Criteria/Marking Scheme:

|                                                       |                 |
|-------------------------------------------------------|-----------------|
| <b>Requirement 1- Report based on report template</b> | <b>50 marks</b> |
|-------------------------------------------------------|-----------------|

|                                                                                                                                                                                                             |                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Write a report which covers the requirements of the whole assignment and submit in turnitin link with properly referenced in Harvard Referencing style.                                                     |                 |
| 1.1 Identify and critically analyse the system architecture, design patterns and frameworks for Enterprise Application architecture. This satisfies the requirements of LO1 partially.                      | 10 marks        |
| 1.2 Critically evaluate the software technologies for building an enterprise application along with their security implications. This satisfies the requirements of LO2 partially.                          | 8 marks         |
| 1.3 Research and document the Business requirements. This satisfies the requirements of LO4 partially.                                                                                                      | 5 marks         |
| 1.4 Make a project plan along with the resources and time based on the requirements and technology selected for the case study. This satisfies the requirements of LO3 partially.                           | 5 marks         |
| 1.5 Document the system design with the system architecture and design diagrams such as UML diagrams, DFD Level 0 and ER diagrams with their explanation. This satisfies the requirements of LO5 partially. | 10 marks        |
| 1.6 Implementation evidence as screenshots with descriptions.                                                                                                                                               | 5 marks         |
| 1.7 Test the results and evaluate how it matches the business requirements. This satisfies the requirements of LO5 partially.                                                                               | 5 marks         |
| 1.8 Recommendation of improvements and security concerns. This satisfies the requirements of LO5 partially.                                                                                                 | 2 marks         |
| <b>Requirement 2 – Software</b>                                                                                                                                                                             | <b>30 marks</b> |
| Develop the software as per the requirements and system design. Submit the code as a Zipped folder with a Readme file. This satisfies the requirements of LO5 partially.                                    |                 |
| 3.1 Software development.                                                                                                                                                                                   | 25 marks        |
| 3.3 Results satisfying all the requirements                                                                                                                                                                 | 5 marks         |
| <b>Requirement 3 – presentation (10 minutes)</b>                                                                                                                                                            | <b>20 marks</b> |

|                                                                                                                                             |         |
|---------------------------------------------------------------------------------------------------------------------------------------------|---------|
| A 10-minute explanatory presentation of the evaluation of the architecture, software development and results.                               |         |
| 4.1 Presentation of the enterprise application architecture and frameworks. This satisfies the requirements of LO1 partially.               | 4 marks |
| 4.2 Presentation of the software technologies for the enterprise system development. This satisfies the requirements of LO2 partially.      | 3 marks |
| 4.3 Presentation of plan and software development process. This satisfies the requirements of LO3 partially.                                | 3 marks |
| 4.4 Critically evaluate the relationship between business goals and software development. This satisfies the requirements of LO4 partially. | 3 marks |
| 4.5 Demonstration of the implemented software project. This satisfies the requirements of LO5 partially.                                    | 5 marks |
| 4.6 Recommendation for security implications. This satisfies the requirements of LO5 partially.                                             | 2 marks |

### **Deliverables**

This assignment is worth 100% of the module mark.

This assignment portfolio should consist of the following:

- The report which satisfies the report template (provided in page 8 – page 11) on your critical evaluation of the software architecture, frameworks. Also the project plan, requirements, design, development, testing and recommendations in turnitin link.
- Weekly progress evidence to the tutor as portfolio.
- Database files and Software project with a Readme File as a Zipped folder(named as StudentNumber\_SoftwareCode).
- Presentation in PDF and the recordings.

### **Report Format :**

#### **1. Introduction to Enterprise System Development**

1.1 Enterprise system architectures Frameworks definition , what are different framework, short description about 2 major framework

1.2 Software Architecture Patterns -3-4

1.3 Selection of architectures frameworks and Pattern

## 1.4 Justification

## 2. Requirements Analysis

### 2.1 Functional requirements

### 2.2 Non-Functional requirements

### 2.3 Mapping the requirements based on business goals

### 2.4 Software and Hardware

## 3. Project Plan

### 3.1 Gantt chart

## 4. System Design

### 4.1 System architecture

### 4.2 System Design

### 4.3 Database Design - ER diagram

### 4.4 Use case diagram

### 4.5 DFD

## 5. Implementation

### 5.1 Screenshots with description as evidence to show Relationship between business goals and software development

## 6. Testing

### 6.1 Test cases as software testing evidence

## 7. Security Implications

## 8. Conclusion

## 9. References

### **Specific Assessment Criteria:**

#### **First class (70% and above):**

Students will provide an in-depth evaluation of the enterprise system development with the enterprise application architecture, frameworks and patterns. Software technologies and hardware required for the enterprise development. Detailed documentation with focused and justified recommendations will be made. Software planning, design and development meet the business requirements in an excellent manner. Extensive research demonstrating use of a wide range of current secondary research sources will be evident. Academic style and referencing will be excellent.

#### **Upper Second Class 2(1) (60-69%):**



Students will provide a comprehensive evaluation of the enterprise system development with the enterprise application architecture, frameworks and patterns. Software technologies and hardware required for the enterprise development. Detailed documentation with justified recommendations will be made. Software planning, design and development meet the business requirements in very good quality. Research demonstrating use of a wide range of current secondary research sources will be evident. Academic style and referencing will be good.

**Lower Second Class 2(2) (50-59%):**

Students will provide a broad evaluation of the enterprise system development with the enterprise application architecture, frameworks and patterns. Software technologies and hardware required for the enterprise development. Detailed documentation with justified recommendations will be made. Software planning, design and development meet the business requirements in good quality. Research demonstrating use of a wide range of current secondary research sources will be evident. Academic style and referencing will be good.

**Third class (40-49%):**

Students will provide a satisfactory evaluation of the enterprise system development with the enterprise application architecture, frameworks and patterns. Software technologies and hardware required for the enterprise development. Detailed documentation with justified recommendations will be made. Software planning, design and development meet the business requirements in satisfactory manner. Research demonstrating use of a range of current secondary research sources will be evident. Academic style and referencing will be fair.

**Fail (39% and below):** Students who do not meet the requirements of a third class grade will not successfully complete the assessment activity.

(Also see Section 17: General Assessment Criteria for Written Assessments)

## 17. General Assessment Criteria for Written Assessments

### GENERAL ASSESSMENT GUIDELINES – LEVEL HE6

|                                                       | <b>Relevance</b><br>Learning outcomes must be met for an overall pass                                                    | <b>Knowledge and Understanding</b>                                                                                                                                                                                                           | <b>Analysis, Creativity and Problem-Solving</b>                                                                                                                                                                                                                                                                                                                                                                                       | <b>Self-awareness and Reflection</b>                                                                                            | <b>Research/ Referencing</b>                                                                                                                                                                 | <b>Written English</b>                                                                                                                                                                                                                                                                                                                                               | <b>Presentation and Structure</b>                                                                                                                                                                                         |
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| Class I<br>(Exceptional Quality)<br><b>85% - 100%</b> | Work is directly relevant and expertly addresses the requirements of the brief.<br><br><b>Learning outcomes are met.</b> | Demonstrates an exceptional breadth and depth of knowledge and understanding of theory and practice beyond the threshold expectation for the level.<br><br>Demonstrates mastery in conceptual understanding of a range of specialised areas. | Presents an exceptional synthesis and critical evaluation of findings from a broad range of relevant sources in order to draw clear, systematic, justified and insightful conclusions.<br>Provides a sophisticated critical insight and expertly interprets complex matters and ideas.<br>Demonstrates exceptional creative flair and a high level of originality.<br>Demonstrates exceptional problem-solving skills and initiative. | Provides insightful reflection and critical self-awareness in relation to the outcomes of own work and personal responsibility. | An extensive range of contemporary and relevant reference sources selected and drawn upon.<br><br>Sources cited accurately in both the body of text and in the Reference List/ Bibliography. | Writing style is clear, succinct and appropriate to the requirements of the assessment. An exceptionally well written answer with competent spelling, grammar and punctuation. For example, paragraphs are well structured and include linking and signposting. Sentences are complete and different types are used. A wide range of appropriate vocabulary is used. | The presentational style and layout are correct for the type of assignment. Evidence of planning and logically structured. Where relevant, there is effective inclusion of, and reference to, figures, tables and images. |
| Class I<br>(Excellent Quality)<br><b>70% - 84%</b>    | Work is relevant and comprehensively addresses the requirements of the brief.<br><br><b>Learning outcomes are met.</b>   | Demonstrates an excellent breadth and depth of knowledge and understanding of theory and practice for this level.<br><br>Demonstrates an in-depth conceptual understanding of a range of specialised areas.                                  | Presents an excellent synthesis and critical evaluation of findings from a broad range of relevant sources in order to draw clear, systematic, justified and perceptive conclusions.<br>Provides a critical insight and clearly interprets complex matters and ideas.<br>Demonstrates creative flair and a high level of originality.<br>Demonstrates excellent problem-solving skills and initiative.                                | Provides excellent reflection and critical self-awareness in relation to the outcomes of own work and personal responsibility.  | A wide range of contemporary and relevant reference sources selected and drawn upon.<br><br>Sources cited accurately in both the body of text and in the Reference List/ Bibliography.       | Writing style is clear, succinct and appropriate to the requirements of the assessment. An excellently well written answer with competent, spelling, grammar and punctuation. For example, paragraphs are well structured and include linking and signposting. Sentences are complete and different types are used. A wide range of appropriate vocabulary is used.  | The presentational style and layout are correct for the type of assignment. Evidence of planning and logically structured. Where relevant, there is effective inclusion of, and reference to, figures, tables and images. |

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| Class II/i<br>(Very Good Quality)<br><b>60% - 69%</b>   | Work is relevant and addresses most of the requirements of the brief well.<br><br><b>Learning outcomes are met.</b>                                      | Demonstrates a thorough breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates a sophisticated conceptual understanding of a range of specialised areas. | Presents a perceptive synthesis and critical evaluation of findings from a range of relevant sources in order to draw clear, justified and thoughtful conclusions. Interprets complex matters and ideas well. Demonstrates a good level of creativity and originality. Demonstrates strong problem-solving skills. | Provides very good reflection and critical self-awareness in relation to the outcomes of own work and personal responsibility, as required by the assessment. | A wide range of relevant reference sources selected and drawn upon.<br><br>Sources cited accurately in the main in both the body of text and in the Reference List/Bibliography. | Writing style is clear, succinct and appropriate to the requirements of the assessment. A very well written answer with competent spelling, grammar and punctuation. For example, paragraphs are well structured and include linking and signposting. Sentences are complete and different types are used. A range of appropriate vocabulary is used. | The presentational style and layout are correct for the type of assignment. Evidence of planning and logically structured in the main. Where relevant, there is effective inclusion of, and reference to, figures, tables and images. |
|                                                         | <b>Relevance</b><br><b>Learning outcomes must be met for an overall pass</b>                                                                             | <b>Knowledge and Understanding</b>                                                                                                                                                                     | <b>Analysis, Creativity and Problem-Solving</b>                                                                                                                                                                                                                                                                    | <b>Self-awareness and Reflection</b>                                                                                                                          | <b>Research/ Referencing</b>                                                                                                                                                     | <b>Written English</b>                                                                                                                                                                                                                                                                                                                                | <b>Presentation and Structure</b>                                                                                                                                                                                                     |
| Class II/ii<br>(Good Quality)<br><b>50% - 59%</b>       | Work addresses key requirements of the brief. Some irrelevant content.<br><br><b>Learning outcomes are met.</b>                                          | Demonstrates a sound breadth and depth of knowledge and understanding of theory and practice for this level.<br><br>Demonstrates a sound conceptual understanding of specialised areas.                | Presents a logical evaluation of findings from a range of relevant sources in order to draw clear and justified conclusions. Interprets some complex matters and ideas. Demonstrates some creativity. Demonstrates effective problem-solving skills and initiative.                                                | Provides good reflection and critical self-awareness in relation to the outcomes of own work and personal responsibility, as required by the assessment.      | A range of relevant reference sources selected and drawn upon.<br><br>Most sources accurately cited both the body of text and in the Reference List/Bibliography.                | Writing style is mostly appropriate to the requirements of the assessment. Grammar, spelling and punctuation are generally competent and minor lapses do not pose difficulty for the reader. Paragraphs are structured and include some linking and signposting. Sentences are complete. A range of appropriate vocabulary is used.                   | The presentational style and layout are correct for the type of assignment. Logically structured in the most part. Inclusion of figures, tables and images but not all relevant or referred to.                                       |
| Class III<br>(Satisfactory Quality)<br><b>40% - 49%</b> | Work addresses the requirements of the brief, although superficially in places. Some irrelevant content.<br><br><b>Learning outcomes are met.</b>        | Demonstrates a sufficient breadth and depth of knowledge and understanding of theory and practice for this level.<br><br>Demonstrates a conceptual understanding of some specialised areas.            | Presents an evaluation of findings from a range of sources in order to draw some valid conclusions. Interprets some complex matters and ideas but with descriptive passages evident which lack clear purpose. Demonstrates creativity in places. Demonstrates sufficient problem-solving skills and initiative.    | Provides some reflection and critical self-awareness in relation to the outcomes of own work and personal responsibility, as required by the assessment.      | Some relevant reference sources selected and drawn upon.<br><br>Some weaknesses in referencing technique.                                                                        | Writing style is occasionally not appropriate for the assessment. Grammar, spelling and punctuation are generally competent, but may pose minor difficulties for the reader. Some paragraphs may lack structure, and there is limited linking and signposting. Some appropriate vocabulary is used                                                    | The presentational style and layout are largely correct for the type of assignment. Adequately structured. Inclusion of some figures, tables and images but not all clear, relevant and/or referred to.                               |
| Borderline Fail<br><b>35% - 39%</b>                     | Work addresses some of the requirements of the brief. Irrelevant and superficial content.<br><br><b>One or more learning outcomes have not been met.</b> | Demonstrates a lack of knowledge and understanding of theory and practice for this level. Demonstrates Insufficient conceptual understanding of specialised areas.                                     | Presents a limited evaluation of findings from set sources. Descriptive or narrative passages evident which lack clear purpose. Demonstrates little creativity. Demonstrates insufficient problem-solving skills and initiative.                                                                                   | Provides limited reflection and critical self-awareness in relation to the outcomes of own work and personal responsibility, when required.                   | Sources selected are limited and lack validity/relevance.<br><br>Poor referencing technique employed.                                                                            | Writing style is unclear and does not match the requirements of the assessment. Deficiencies in spelling, grammar and punctuation makes reading difficult and arguments unclear in places. Paragraphs are poorly structured.                                                                                                                          | For the type of assignment the presentational style, layout and/or structure are lacking. Inclusion of figures, tables and images but not clear, relevant and/or referred to.                                                         |

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| <p>Fail<br/>&lt;34%</p> | <p>Work does not address the requirements of the brief. Irrelevant and superficial content.</p> <p><b>One or more learning outcomes have not been met.</b></p> | <p>Demonstrates inadequate knowledge and understanding of theory and practice for this level.</p> <p>Demonstrates Insufficient conceptual understanding of relevant areas.</p> | <p>Analysis is weak and poorly constructed with inadequate sources drawn upon. Demonstrates little or no creativity.</p> <p>Demonstrates a lack of problem-solving skills and initiative.</p> | <p>Provides inadequate reflection and self-awareness in relation to the outcomes of own work and personal responsibility, when required.</p> | <p>An absence of relevant sources selected and drawn upon.</p> <p>Poor referencing technique employed.</p> | <p>Writing style is unclear and does not match the requirements of the assessment in question. Deficiencies in spelling, grammar and punctuation makes reading difficult and arguments unclear. Unstructured paragraphs.</p> | <p>For the type of assignment the presentational style, layout and/or structure are lacking.</p> <p>Inclusion of figures, tables and images but not clear, relevant and/or referred to.</p> |
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